

## Installing Nipper

Nipper is available for both the Linux and Windows platform. It can be downloaded from [nipper.titania.co.uk/downloads.php](http://nipper.titania.co.uk/downloads.php). For Windows, it's quite straightforward. Download the zip file containing the Nipper executable, extract all contents and use the command prompt to execute *nipper.exe*.

Installing it on Linux is also simple. Two packages are required on Linux: *libnipper* and *nipper-cli*. Download the source code and follow the steps mentioned on [nipper.titania.co.uk/install.php](http://nipper.titania.co.uk/install.php). The steps to install from source:

1. Download and extract.
2. Create a *build* directory in the source code directory
3. Run the *cmake* command in the *build* directory
4. Run *make* and *make install*

Alternatively, you can use the BackTrack Live CD to use Nipper without the requirement for a local installation.

## Getting started

To begin with, you need to provide the network device configuration file as the input to Nipper. The tool then parses this file and analyses each Access Control List (ACL) and Access Control Entries (ACE). Shown below is the sample output displaying Nipper running on the BackTrack LiveCD:

```
bt nipper # nipper -ios-router -input=router.conf -output=ciscoreport.html
bt nipper #
```

Nipper is capable of exporting reports in HTML (default), XML, Latex and ASCII text report formats. Figure

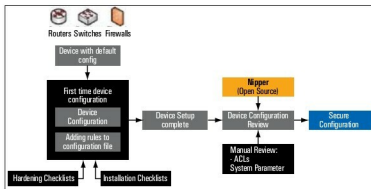


Figure 4: Best practices for a network device review

2 shows a sample of an HTML report output. Command line options for selecting a report format are as follows:

- *--html* for an HTML report format
- *--xml* for an XML report format
- *--latex* for a LaTeX report format
- *--text* for a typical ASCII text report format

As you're aware by now, Nipper exists to report vulnerabilities in network configurations. As part of this reporting, Nipper includes the following sections: observations made during audit, the impact of a security weakness, ease of carrying out the attack, and recommendations to remove the security weaknesses. A sample report is shown in Figure 3.

## Best practices

Before we close for now, here are some of the best practices that should be followed in your current IT set-up to ensure a secure network perimeter:

1. Develop a security/hardening checklist for each vendor product. Ensure that it is referred to before a device is installed.
2. Device roll-out to production must happen through a change/IT request. The security team should validate device hardening with Nipper, while going through a manual review of ACLs for that particular set-up.
3. Track security notifications for network products that are installed on the network.
4. Network security reviews and audits must be conducted on a frequent basis by external network security consultants.

Figure 4 represents the best practices for a network review. Of course, there's more to be said here, but I'll leave it to you to test Nipper and find out how suitable it is for your network auditing requirements. Point your browser to [nipper.titania.co.uk](http://nipper.titania.co.uk) to get started. 

By: **Rajinder Singh**

The author is an information security consultant at TCS. A penetration tester by profession, he has been using Linux since 2002.

Table 1

| Device and command switches                  |  |                |
|----------------------------------------------|--|----------------|
| Device Type                                  |  | CLI Option     |
| Checkpoint VPN-1/Firewall-1                  |  | --fw1          |
| Bay Networks Accelar                         |  | --accelar      |
| Cisco Catalysts (IOS)                        |  | --ios-catalyst |
| Cisco Catalyst (NMP/CatOS):                  |  |                |
| i) NMP based catalyst                        |  | --nmp          |
| ii) CatOS based catalyst                     |  | --catos        |
| Cisco CSS                                    |  | --css          |
| Cisco Routers (IOS)                          |  | --ios-router   |
| Cisco Security Appliances (ASA / PIX / FWSM) |  |                |
| i) Cisco ASA Security Appliance              |  | --asa          |
| ii) Cisco PIX Security Appliance             |  | --pix          |
| iii) Cisco FWSM Security Appliance           |  | --fwsn         |
| Juniper NetScreen Firewalls                  |  | --screenos     |
| Nokia IP Firewalls                           |  | --nokia        |
| Nortel Passport                              |  | --passport     |
| SonicWALL SonicOS Firewall                   |  | --sonicos      |